

CSC 360, Github

Introduction

Github (<https://github.com>) is a development platform and is probably the most popular web-based source code management system. Github uses *git* for revision control and source code management. In addition, Github implements collaborative features such as bug tracking and feature requests. Continuous integration (CI), which includes automated testing, is possible using Github workflows or third party services. Github is free to use (additional features are available for an additional charge). Other source code management systems such as BitBucket (<https://bitbucket.org>) are available as well. For Github, both command line *git* tools and a GUI tool called Github Desktop are available. Github is also integrated into IDEs such as Visual Studio Code.

Why use Github?

- Github is a version control system:
 - o it leaves an audit trail of all code changes (human and/or AI-generated!)
 - o you can easily revert to a previous version of code, if needed
 - o you can easily see how files change from one version to another
 - o branches can be used to separate major versions of code (e.g., release and development versions; see figure below)
- Github makes it easy to work with existing code and to collaborate:
 - o a user can (a) *fork* (copy) code from anyone's repository, (b) clone the code (i.e. make a local copy for development), (c) make changes to the code, (d) commit (record) the changes and (e) push (save) the changes to the remote repository (i.e., the repository on github.com).
 - o if appropriate, the user can submit a *pull request*, which is a request to merge his/her code with another branch or version, possibly from another user
 - o pull requests can be automatically merged if there are no conflicts; otherwise conflicts are noted and must be manually resolved.
- Github makes it easy to automate workflows, such as testing and deployment, through Github actions (<https://docs.github.com/en/actions>)



Figure 1. Example Github branches, source: <http://nvie.com/posts/a-successful-git-branching-model/>

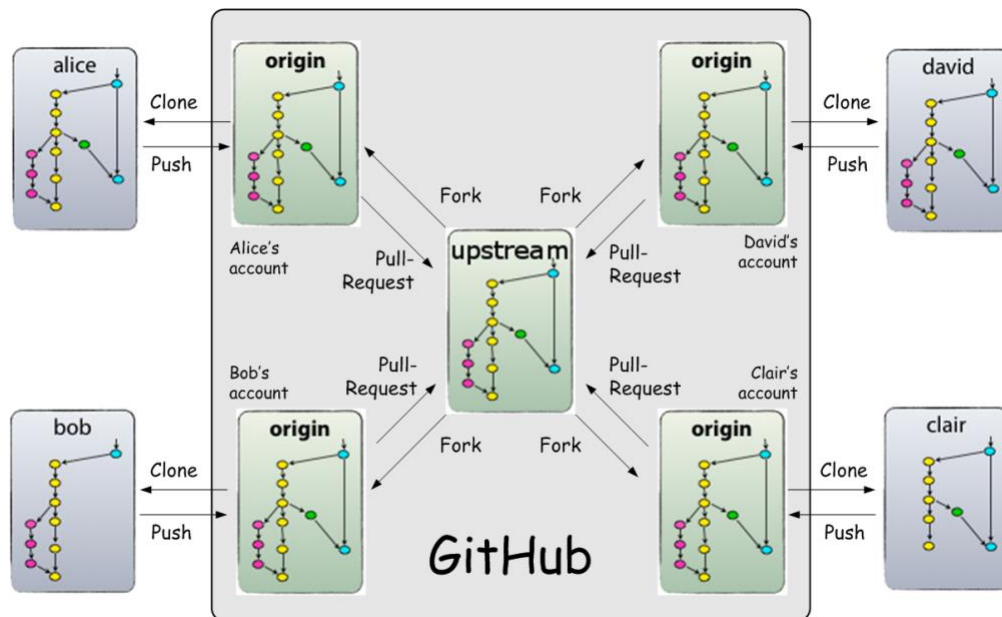


Figure 2. Collaboration on Github, source: <http://www.dalescott.net/2012/09/14/using-gitflow-with-githubs-fork-pull-model/>

Let's look at an example, <https://github.com/gdancik/CoursePlanningTool>

Github cheatsheet: <https://education.github.com/git-cheat-sheet-education.pdf>